
Lossfunk Autoresearch Submission

Hormetic IB Scheduling for Persistent Object Identity

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github.com/Incharajayaram/hormetic-ib-slot

Tool: Autovoila (Lossfunk) + Claude Code (claude-sonnet-4-6)

1. Starting Research Question

The Claim

Slot-based object-centric representations trained under a **hormetic** Information-Bottleneck schedule (beta increasing progressively during training) retain object identity through occlusion more robustly than identical architectures trained at fixed beta or non-monotone beta schedules.

The Intervention is Minimal

Same architecture, same parameter count, same final beta

Only the beta(t) trajectory differs across 6 conditions

Primary metric: identity-retention accuracy under occlusion on CLEVRER and ADEPT

One-Sentence Version

A slot-based object representation trained under a hormetic IB schedule maintains object identity through occlusion more robustly than fixed-beta or non-monotone-beta baselines, measurable on CLEVRER and ADEPT against published slot-based persistence systems.

2. Autoresearch Flow

Tool

Autovoila (Lossfunk prompt) on Claude Code (claude-sonnet-4-6), YOLO mode via cco.

Session Flow

Provided the Research Question Sharpener as starting prompt

Claude scaffolded the full project in one session: Slot Attention encoder, VIB head, 6 schedule classes, training/eval pipeline, data loaders

Ran a synthetic pilot (toy Gaussian data, CPU, 100 steps) to verify the pipeline

No real CLEVRER/ADEPT runs — require A100 + ~450 GPU-hours

Customization

None to the Autovoila prompt

Research proposal was detailed enough that Claude needed very little mid-session direction

Outputs reviewed but not edited

Human Intervention Level: Low

Provided initial framing (Research Question Sharpener), reviewed output, did not touch artifact code or results.

3. Summary of Results

All numbers from synthetic experiments (CPU only). Not from CLEVRER or ADEPT. 3 seeds x 300 steps; mean shown.

Condition	IRA @k=2	IRA @k=4	IRA @k=6	Slot Collapse
Hormetic-Sigmoid	0.248	0.254	0.252	39.6%
Hormetic-Cosine	0.291	0.277	0.304	62.6%
Linear	0.283	0.237	0.233	50.0%
Reverse	0.335	0.242	0.185	53.7%
Random Permutation	0.314	0.281	0.285	50.9%
Fixed-beta (baseline)	0.318	0.227	0.207	70.4%

Collapse finding: Hormetic-Sigmoid has the lowest collapse (39.6%) vs Fixed-beta (70.4%). Directionally consistent with the hypothesis mechanism.

IRA finding: No condition dominates clearly on identity retention. High variance (std often 50-100% of mean) means 3 seeds is insufficient for statistically meaningful comparison.

4. What I Learned About the Research Question

What the AI Confirmed

The experimental design is correct and the controls are well-chosen

The code runs; the question can be operationalized

The scaffold is a genuine starting point for real experiments

What Surprised Me

The question is easier to produce *results that look like evidence* without being evidence than I expected. The numerical ordering on synthetic data matches the hypothesis directionally, but that synthetic data was never validated as a proxy for CLEVRER/ADEPT dynamics. A direction match on toy data is not a result.

The Deeper Gap

The research question is well-posed at the proposal level, but the meaningful empirical content lives entirely in the actual training runs on real data with real occlusion events. That is precisely what the auto-research system could not do. The gap between 'scaffold is correct' and 'experiment ran' is the entire scientific content of this project.

5. Critique: What the AI Got Right

The codebase is structurally clean and correct.

Component	Why it's correct
VIB loss	$L = \text{recon_loss} + \text{beta} \times \text{KL}(q(z x) p(z))$ with closed-form diagonal Gaussian KL
Hungarian matching	Correctly used for slot-to-object assignment in identity-retention evaluation
6-condition ablation	Random permutation + reverse baselines are the right controls to isolate schedule geometry
Identity-retention metric	Same slot index maps to same object pre/post occlusion at k in $\{4,8,16,32\}$ frames
Data loaders	CLEVRER + ADEPT interfaces structurally correct; would work with real datasets present
README	Honestly documents the claim, expected compute (~450 GPU-hours), and data requirements

6. Critique: What the Original AI Run Got Wrong

1. The experiments are not the experiments.

Synthetic Gaussian data is not CLEVRER or ADEPT. The claim is about visual occlusion in video. The AI ran what it could and reported it as informative. It is not.

2. Eval bug: IRA @k=4 was always 0.0.

Off-by-one in the original evaluation: 'if $T > k$ ' with $T=4$, $k=4$ silently skipped the $k=4$ bucket, returning 0.0. This was not flagged as anomalous. Fixed by changing to 'if $T \geq k$ ' and $T=6$ to match config.

3. 75% slot collapse with no diversity penalty.

The original run had no mechanism to prevent slot collapse. When slots degenerate, IRA measures noise routing. Fixed by adding a slot diversity auxiliary loss ($\lambda=0.05$).

4. Single seed, 100 steps: not enough signal.

One seed at 100 steps on toy data gives zero statistical power. The IRA ordering at $k=2$ (Hormetic-Cosine first) was likely noise. Fixed by running 3 seeds at 300 steps with aggregated mean $\pm$ std.

5. The paper was never written.

The caisc_2026.tex in the original repo is the blank template. No abstract, no methods, no results in paper form. A full CAISc 2026 draft has now been written.

6. No IB-plane trajectories, no OOD evaluation.

Both listed as secondary measurements in the design. Neither was produced, even on the synthetic dataset.

7. Reflection: Limits of Auto-Research on This Question

What the AI did well

Translated a well-specified proposal into working code fast; a human needs 2-4 weeks for the same scaffold

Picked the right components: VIB, Hungarian matching, SAVi-style temporal slot attention, spatial broadcast decoder

Designed the correct ablation structure including matched-beta controls

What kind of question would AI do better on?

A question where compute is small, data is accessible, evaluation is unambiguous, and prior literature has a clear quantitative baseline to beat. Something like: 'Does this architectural change improve ARI on CLEVRER in under 10 GPU-hours?' That is nearly fully automatable. This question requires 450 GPU-hours, two datasets, and evaluation that depends on ruling out multiple confounds.

What the AI failed at

Cannot run experiments requiring GPUs and datasets it does not have. The entire empirical content of the claim lives in 450 GPU-hours on CLEVRER/ADEPT.

Did not recognize when its own results were invalid in the first pass. After prompting, diagnosed the eval bug ($k=4$ always 0.0), identified slot collapse as the root cause, and corrected both in a second run.

'The pipeline ran' is not 'the experiment worked.' Running without error on toy data is not evidence.

Cannot construct the interpretive argument. The novel claim requires reading IB theory, Slot Attention, and Saxe et al.'s critique together with nuance the AI cannot provide.

8. Revised Research Plan

What Stays the Same

Core claim is still worth testing; experimental design is correct
AI-generated codebase is a usable starting point

What Was Fixed in This Session

Eval bug found and fixed: IRA @k=4 now non-zero across all conditions
Slot diversity loss added ($\lambda=0.05$): hormetic_sigmoid collapse drops from 75% to 39.6%; fixed_beta remains at 70.4%
3-seed 300-step ablation run: results have error bars, not just point estimates
Full CAISc 2026 paper draft written with real numbers filled in

What Still Cannot Be Done Without GPU + Data

Experiments on CLEVRER or ADEPT (approx 450 GPU-hours on A100)
ARI evaluation on real video frames
IB information-plane trajectories at scale

What the Corrected Results Show

The collapse finding is directionally consistent: hormetic_sigmoid has the lowest collapse (39.6%) and fixed_beta has the highest (70.4%). This supports the mechanism hypothesis that progressive compression prevents early slot degeneration.

What the Corrected Results Do Not Show

The IRA numbers do not cleanly favor hormetic conditions. No single condition dominates across $k=2,4,6$. Variance is high (std often 50-100% of mean). 3 seeds on CPU at 300 steps is insufficient for statistically meaningful schedule comparison. The collapse finding is the only signal worth reporting.

9. Summary

Item	Status
Research question	Well-posed, falsifiable, still worth testing
Codebase / scaffold	AI-generated, structurally correct, usable
Experiments (CLEVRER/ADEPT)	Not run — require GPU and real datasets
Synthetic pilot	Corrected: eval bug fixed, 3-seed 300-step ablation, diversity loss added
Paper draft	Written: CAISc 2026 draft with corrected numbers
GitHub repo	github.com/Incharajayaram/hormetic-ib-slot

The honest one-sentence version:

The AI built the lab but could not run the experiment, and did not recognize that the dry-run it did run had failed.

Generated with Autovoila (Lossfunk) + Claude Code. Human contribution: research proposal, Research Question Sharpener, this critique.